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# **EECS 348: Software Engineering I**

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**EECS 348 Group 5**

**Test Case**

**Version <1.6>**

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| <EECS 348 Group 5> | Version: <1.6>     |
| Test Case          | Date: <05/07/2026> |

## Revision History

| Date     | Version | Description                                                           | Author      |
|----------|---------|-----------------------------------------------------------------------|-------------|
| 5/4/2026 | 1.1     | Completed Test Cases for Valid Expressions 1 - 4                      | Lea         |
| 5/4/2026 | 1.2     | Completed Test Cases for Valid Expressions 5 - 8                      | Jr          |
| 5/4/2026 | 1.3     | Completed Test Cases for Valid Expressions 9 - 12                     | Alex        |
| 5/4/2026 | 1.4     | Completed Test Cases for Valid Expressions 13 - 15                    | Zack        |
| 5/4/2026 | 1.5     | Completed Test Cases for Invalid Expressions 1 - 5                    | Shashwat    |
| 5/5/2026 | 1.6     | Completed Document for Purpose Section, and Invalid Test Cases 6 - 10 | Alec Thomas |

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## Test Cases

### 1. Purpose

This Test Case Specifications document defines the test cases for the *Arithmetic Expression Evaluator*. The purpose of this document is to specify the inputs, expected outputs, and test conditions used to verify that the program correctly parses and evaluates arithmetic expressions according to PEMDAS rules, properly handles parentheses and unary operators, and robustly detects and reports errors for invalid input.

The test cases are organized into two main categories:

- Valid Expressions - to confirm correct evaluation of expressions involving numeric constants and the supported operators (+, -, \*, /, \*\*, %), including proper precedence and associativity.
- Invalid Expressions - to ensure the program correctly identifies and reports errors such as unmatched parentheses, division by zero, missing operands, invalid operator sequences, and invalid characters.

These test cases support the verification of the implementation against the project requirements.

### 2. Test Cases

#### Valid Expressions

|                       |                                     |
|-----------------------|-------------------------------------|
| Test Case Identifier  | TC-VAL-01                           |
| Test Item             | Basic Addition of Numeric Constants |
| Input Specifications  | 3 + 4                               |
| Output Specifications | 7                                   |
| Actual Result         | 7                                   |
| Status                | PASS                                |

|                       |                                          |
|-----------------------|------------------------------------------|
| Test Case Identifier  | TC-VAL-02                                |
| Test Item             | Subtraction Precedence Using Parentheses |
| Input Specifications  | 8 - (5 - 2)                              |
| Output Specifications | 5                                        |
| Actual Result         | 5                                        |
| Status                | PASS                                     |

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|                              |                                                             |
|------------------------------|-------------------------------------------------------------|
| <b>Test Case Identifier</b>  | TC-VAL-03                                                   |
| <b>Test Item</b>             | Left-to-Right Associativity for Multiplication and Division |
| <b>Input Specifications</b>  | $10 * 2 / 5$                                                |
| <b>Output Specifications</b> | 4                                                           |
| <b>Actual Result</b>         | 4                                                           |
| <b>Status</b>                | PASS                                                        |

|                              |                                    |
|------------------------------|------------------------------------|
| <b>Test Case Identifier</b>  | TC-VAL-04                          |
| <b>Test Item</b>             | Exponentiation Operator Evaluation |
| <b>Input Specifications</b>  | $2 ** 3$                           |
| <b>Output Specifications</b> | 8                                  |
| <b>Actual Result</b>         | 8                                  |
| <b>Status</b>                | PASS                               |

|                              |                            |
|------------------------------|----------------------------|
| <b>Test Case Identifier</b>  | TC-VAL-05                  |
| <b>Test Item</b>             | Mixed Operators Evaluation |
| <b>Input Specifications</b>  | $4 * (3 + 2) \% 7 - 1$     |
| <b>Output Specifications</b> | 5                          |
| <b>Actual Result</b>         | 5                          |
| <b>Status</b>                | PASS                       |

|                              |                                                         |
|------------------------------|---------------------------------------------------------|
| <b>Test Case Identifier</b>  | TC-VAL-06                                               |
| <b>Test Item</b>             | Complex Addition with Extraneous Parentheses Evaluation |
| <b>Input Specifications</b>  | $((2 + 3)) + (((1 + 2)))$                               |
| <b>Output Specifications</b> | 8                                                       |
| <b>Actual Result</b>         | 8                                                       |
| <b>Status</b>                | PASS                                                    |

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|                              |                                                        |
|------------------------------|--------------------------------------------------------|
| <b>Test Case Identifier</b>  | TC-VAL-07                                              |
| <b>Test Item</b>             | Mixed Operators with Extraneous Parentheses Evaluation |
| <b>Input Specifications</b>  | $((5 * 2) - ((3 / 1) + ((4 \% 3))))$                   |
| <b>Output Specifications</b> | 6                                                      |
| <b>Actual Result</b>         | 6                                                      |
| <b>Status</b>                | PASS                                                   |

|                              |                                                      |
|------------------------------|------------------------------------------------------|
| <b>Test Case Identifier</b>  | TC-VAL-08                                            |
| <b>Test Item</b>             | Nested Parentheses with Exponents Evaluation         |
| <b>Input Specifications</b>  | $((2 ** (1 + 1)) + ((3 - 1) ** 2)) / ((4 / 2) \% 3)$ |
| <b>Output Specifications</b> | 4                                                    |
| <b>Actual Result</b>         | 4                                                    |
| <b>Status</b>                | PASS                                                 |

|                              |                                                     |
|------------------------------|-----------------------------------------------------|
| <b>Test Case Identifier</b>  | TC-VAL-09                                           |
| <b>Test Item</b>             | Combination of Extraneous and Necessary Parentheses |
| <b>Input Specifications</b>  | $(((((5-3))) * (((2+1)))) + ((2 * 3))))$            |
| <b>Output Specifications</b> | 12                                                  |
| <b>Actual Result</b>         | 12                                                  |
| <b>Status</b>                | PASS                                                |

|                              |                                           |
|------------------------------|-------------------------------------------|
| <b>Test Case Identifier</b>  | TC-VAL-10                                 |
| <b>Test Item</b>             | Extraneous Parentheses with Division      |
| <b>Input Specifications</b>  | $((9 + 6)) / ((3 * 1) / (((2 + 2))) - 1)$ |
| <b>Output Specifications</b> | -60                                       |
| <b>Actual Result</b>         | -60                                       |
| <b>Status</b>                | PASS                                      |

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|                              |                                                       |
|------------------------------|-------------------------------------------------------|
| <b>Test Case Identifier</b>  | TC-VAL-11                                             |
| <b>Test Item</b>             | Combining Unary Operators with Arithmetic Operations: |
| <b>Input Specifications</b>  | $+(-2) * (-3) - ((-4) / (+5))$                        |
| <b>Output Specifications</b> | 6.8                                                   |
| <b>Actual Result</b>         | 6.8                                                   |
| <b>Status</b>                | PASS                                                  |

|                              |                                            |
|------------------------------|--------------------------------------------|
| <b>Test Case Identifier</b>  | TC-VAL-12                                  |
| <b>Test Item</b>             | Unary Negation and Addition in Parentheses |
| <b>Input Specifications</b>  | $- (+1) + (+2)$                            |
| <b>Output Specifications</b> | 1                                          |
| <b>Actual Result</b>         | 1                                          |
| <b>Status</b>                | PASS                                       |

|                              |                                                                                                |
|------------------------------|------------------------------------------------------------------------------------------------|
| <b>Test Case Identifier</b>  | TC-VAL-13                                                                                      |
| <b>Test Item</b>             | demonstrates nested unary negations and additions, with some values negated, and others added. |
| <b>Input Specifications</b>  | $-(-(-3)) + (-4) + (+5)$                                                                       |
| <b>Output Specifications</b> | -2                                                                                             |
| <b>Actual Result</b>         | -2                                                                                             |
| <b>Status</b>                | PASS                                                                                           |

|                              |                                                                                            |
|------------------------------|--------------------------------------------------------------------------------------------|
| <b>Test Case Identifier</b>  | TC-VAL-14                                                                                  |
| <b>Test Item</b>             | The unary + and - operators are used with exponentiation to calculate a fractional result. |
| <b>Input Specifications</b>  | $+2 ** (-3)$                                                                               |
| <b>Output Specifications</b> | 0.125                                                                                      |
| <b>Actual Result</b>         | 0.125                                                                                      |

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| Status | PASS |
|--------|------|

|                       |                                                                                      |
|-----------------------|--------------------------------------------------------------------------------------|
| Test Case Identifier  | TC-VAL-15                                                                            |
| Test Item             | This expression combines unary operators with parentheses and arithmetic operations. |
| Input Specifications  | $-(+2) * (+3) - (-4) / (-5)$                                                         |
| Output Specifications | -6.8                                                                                 |
| Actual Result         | -6.8                                                                                 |
| Status                | PASS                                                                                 |

### Invalid Expressions

|                       |                                    |
|-----------------------|------------------------------------|
| Test Case Identifier  | TC-INV-01                          |
| Test Item             | Unmatched Parentheses              |
| Input Specifications  | $2 * (4 + 3 - 1$                   |
| Output Specifications | Error: Missing closing parenthesis |
| Actual Result         | Error: Missing closing parenthesis |
| Status                | PASS                               |

|                       |                            |
|-----------------------|----------------------------|
| Test Case Identifier  | TC-INV-02                  |
| Test Item             | Operators Without Operands |
| Input Specifications  | $* 5 + 2$                  |
| Output Specifications | Error: Invalid expression  |
| Actual Result         | Error: Invalid expression  |
| Status                | PASS                       |



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|                              |                         |
|------------------------------|-------------------------|
| <b>Test Case Identifier</b>  | TC-INV-03               |
| <b>Test Item</b>             | Division by Zero        |
| <b>Input Specifications</b>  | 4 / 0                   |
| <b>Output Specifications</b> | Error: Division by zero |
| <b>Actual Result</b>         | Error: Division by zero |
| <b>Status</b>                | PASS                    |

|                              |                                                              |
|------------------------------|--------------------------------------------------------------|
| <b>Test Case Identifier</b>  | TC-INV-04                                                    |
| <b>Test Item</b>             | Missing Operator                                             |
| <b>Input Specifications</b>  | 5 (2 + 3)                                                    |
| <b>Output Specifications</b> | Error: Invalid expression: unexpected characters at the end. |
| <b>Actual Result</b>         | Error: Invalid expression: unexpected characters at the end. |
| <b>Status</b>                | PASS                                                         |

|                              |                                                              |
|------------------------------|--------------------------------------------------------------|
| <b>Test Case Identifier</b>  | TC-INV-05                                                    |
| <b>Test Item</b>             | Invalid Characters                                           |
| <b>Input Specifications</b>  | 7 & 3                                                        |
| <b>Output Specifications</b> | Error: Invalid expression: unexpected characters at the end. |
| <b>Actual Result</b>         | Error: Invalid expression: unexpected characters at the end. |
| <b>Status</b>                | PASS                                                         |

|                              |                                    |
|------------------------------|------------------------------------|
| <b>Test Case Identifier</b>  | TC-INV-06                          |
| <b>Test Item</b>             | Unmatched Parenthesis              |
| <b>Input Specifications</b>  | ((3 + 4) - 2) + (1)                |
| <b>Output Specifications</b> | Error: Missing closing parenthesis |
| <b>Actual Result</b>         | Error: Missing closing parenthesis |

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| Status | PASS |
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|                       |                                       |
|-----------------------|---------------------------------------|
| Test Case Identifier  | TC-INV-07                             |
| Test Item             | Invalid Operator Usage/Divide by zero |
| Input Specifications  | $((5 + 2) / (3 * 0))$                 |
| Output Specifications | Error: Division by zero               |
| Actual Result         | Error: Division by zero               |
| Status                | PASS                                  |

|                       |                           |
|-----------------------|---------------------------|
| Test Case Identifier  | TC-INV-08                 |
| Test Item             | Invalid Operator Sequence |
| Input Specifications  | $((2 -) 1 + 3)$           |
| Output Specifications | Error: Invalid expression |
| Actual Result         | Error: Invalid expression |
| Status                | PASS                      |

|                       |                           |
|-----------------------|---------------------------|
| Test Case Identifier  | TC-INV-09                 |
| Test Item             | Missing Operand           |
| Input Specifications  | $((4 * 2) + ( - ))$       |
| Output Specifications | Error: Invalid expression |
| Actual Result         | Error: Invalid expression |
| Status                | PASS                      |

|                       |                                    |
|-----------------------|------------------------------------|
| Test Case Identifier  | TC-INV-10                          |
| Test Item             | Invalid Characters                 |
| Input Specifications  | $((7 * 3) ^ 2)$                    |
| Output Specifications | Error: Missing closing parenthesis |
| Actual Result         | Error: Missing closing parenthesis |
| Status                | PASS                               |

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